

Aedes Mosquito Image Classification

Using EfficientNetV2-S Transfer Learning

Dataset:

Aedes Mosquito Image Dataset Version 1.0 (AMID v1)

Model:

EfficientNetV2-S — ImageNet Transfer Learning

Number of Classes: 4

Total Selected Images: 2000

Dataset Information

Dataset path (root): /content/aedes_dataset/AMID V1

Number of classes: 4

Class names: Aedes_aegypti, Aedes_albopictus, Culex_pipiens, Culex_quinquefasciatus

Images per class: 500

Total images: 2000

Train / Validation / Test split: 1400 / 300 / 300 (70% / 15% / 15%, stratified)

Model Architecture

Base: EfficientNetV2-S, ImageNet pretrained weights, include_top=False

-> GlobalAveragePooling2D

-> BatchNormalization

-> Dropout(0.3)

-> Dense(256, activation='relu')

-> Dropout(0.3)

-> Dense(4, activation='softmax', dtype='float32')

Trainable Parameters: 10,641,284

Non-trainable Parameters: 10,019,040

Total Parameters: 20,660,324

Training Configuration

Optimizer: Adam

Stage-1 learning rate: 0.001

Fine-tuning learning rate (if used): 0.0001

Batch size: 32

Max epochs per stage: 30

Loss: categorical_crossentropy

Augmentations (train only): RandomRotation, HorizontalFlip, ZoomRange, Width/Height Shift, BrightnessRange

EarlyStopping: monitor=val_loss, patience=7, restore_best_weights=True

ReduceLROnPlateau: monitor=val_loss, factor=0.2, patience=3, min_lr=1e-6

ModelCheckpoint: saves best model to results_EfficientNetV2S/EfficientNetV2-S_best.keras

Fine-tuning used: True

Performance Summary

Model Name: EfficientNetV2-S

Training Accuracy: 96.14%

Validation Accuracy: 92.00%

Testing Accuracy: 92.00%

Precision: 92.05%

Recall: 92.00%

F1 Score: 91.93%

Macro F1: 91.93%

Weighted F1: 91.93%

ROC AUC: 99.32%

Trainable Parameters: 10,641,284

Total Parameters: 20,660,324

Training Time: 48.32 min

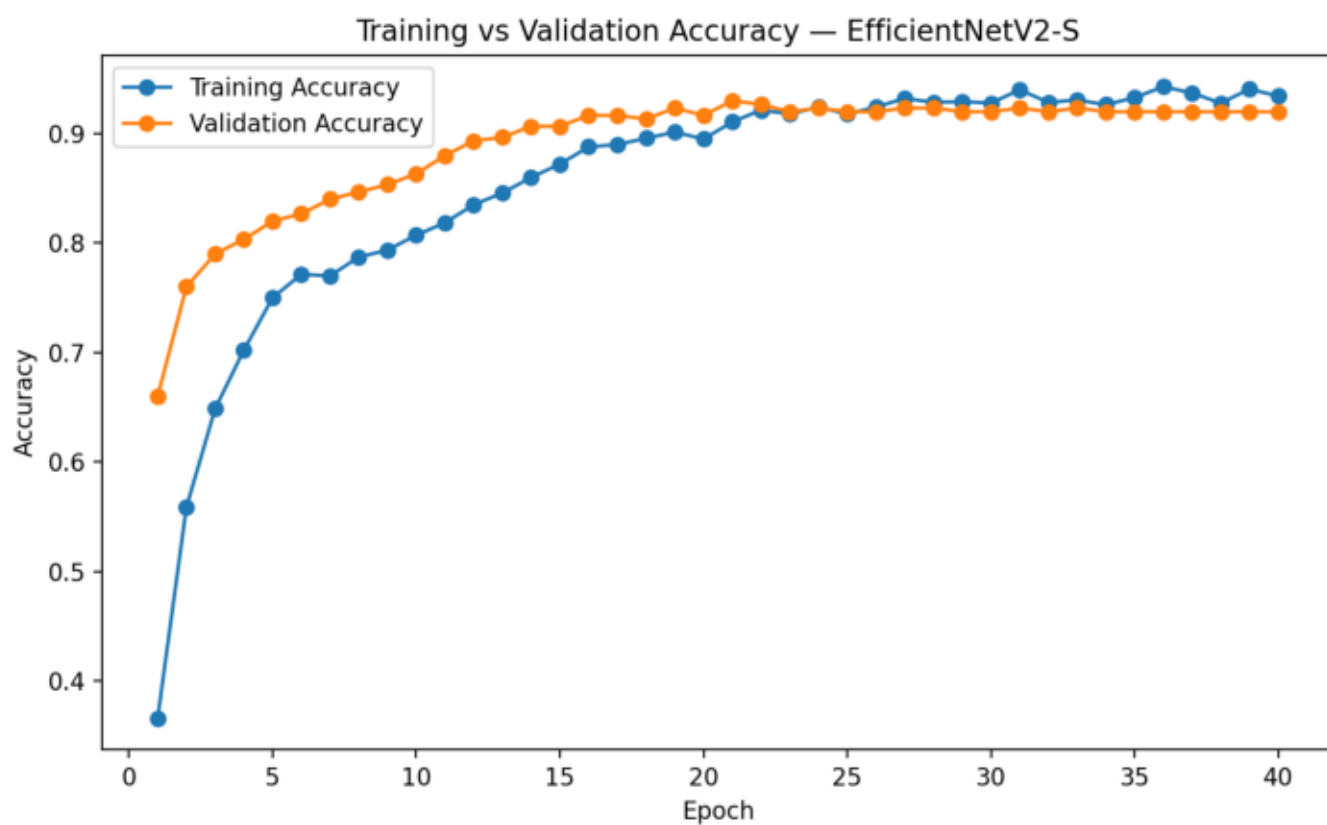
Inference Time: 4.26 ms/image

Fine-Tuning Used: True

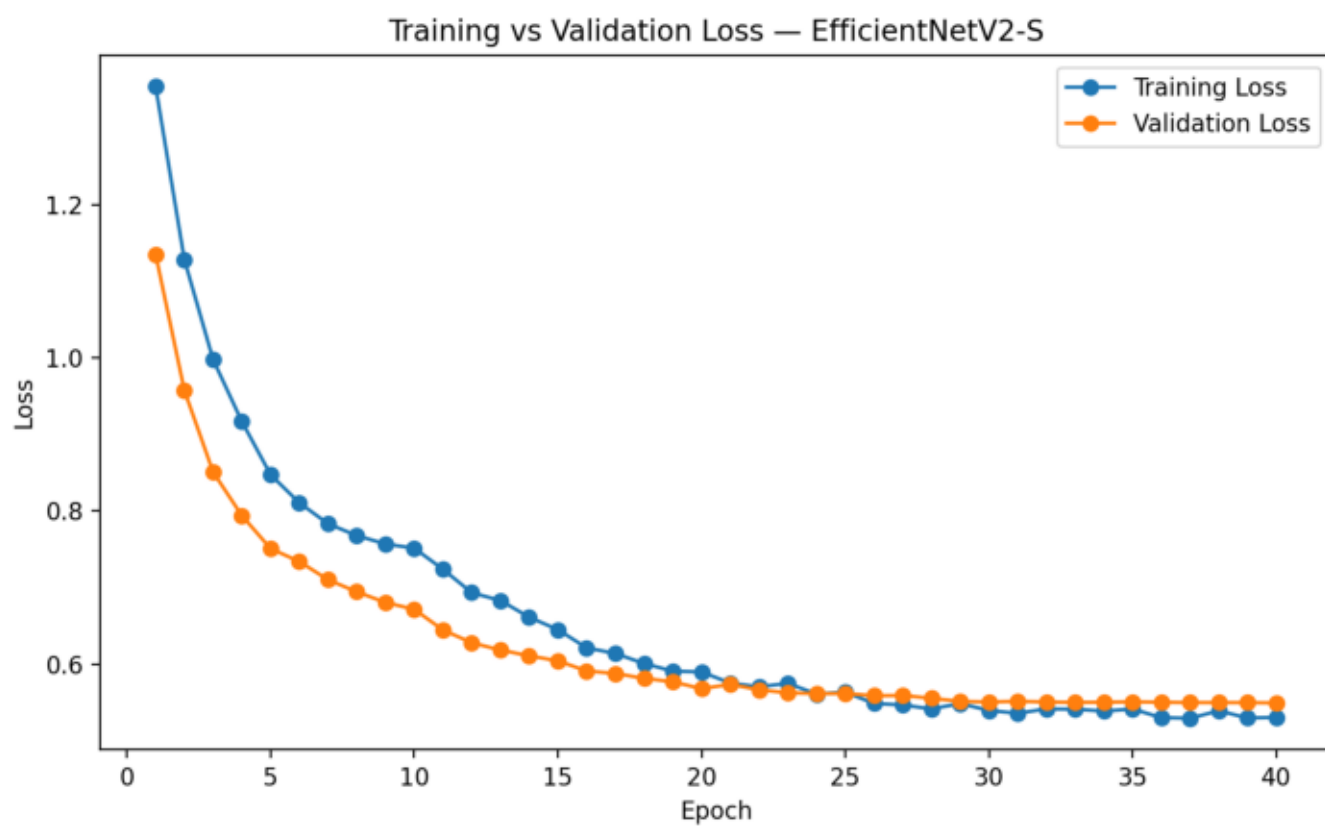
Classification Report

	precision	recall	f1-score	support
Aedes_aegypti	0.9403	0.8400	0.8873	75
Aedes_albopictus	0.8571	0.8800	0.8684	75
Culex_pipiens	0.9231	0.9600	0.9412	75
Culex_quinquefasciatus	0.9615	1.0000	0.9804	75
accuracy			0.9200	300
macro avg	0.9205	0.9200	0.9193	300
weighted avg	0.9205	0.9200	0.9193	300

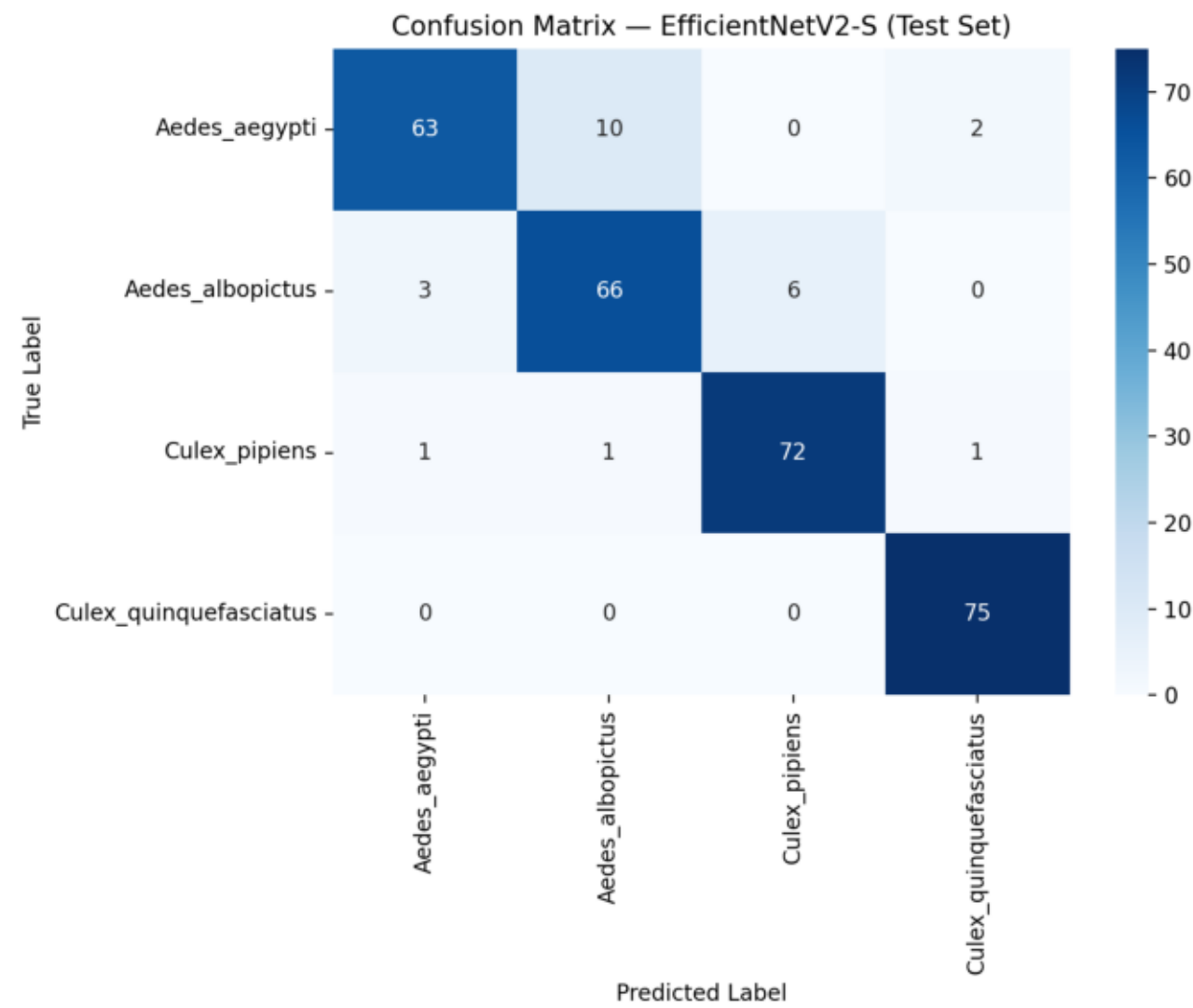
Accuracy Curve



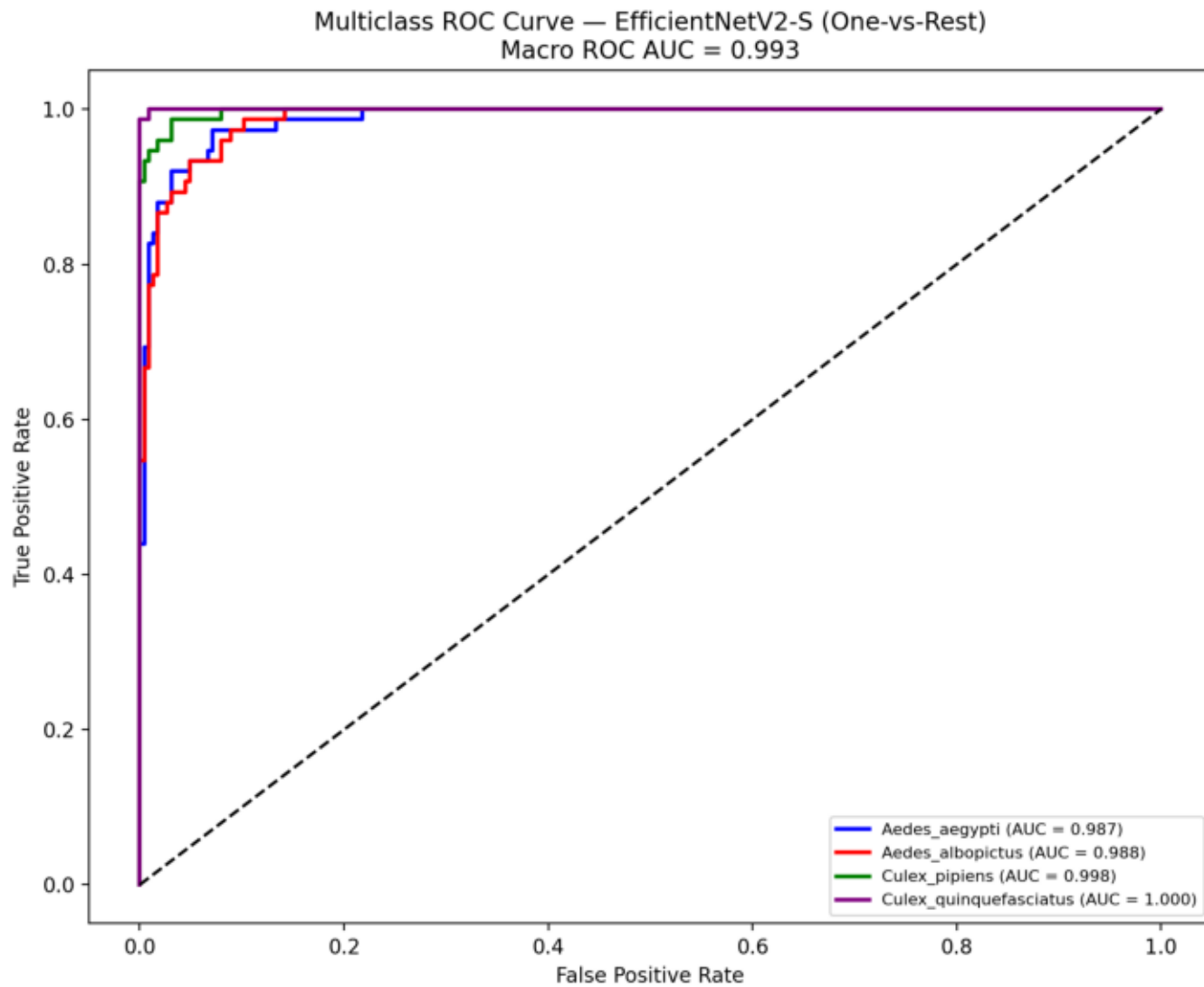
Loss Curve



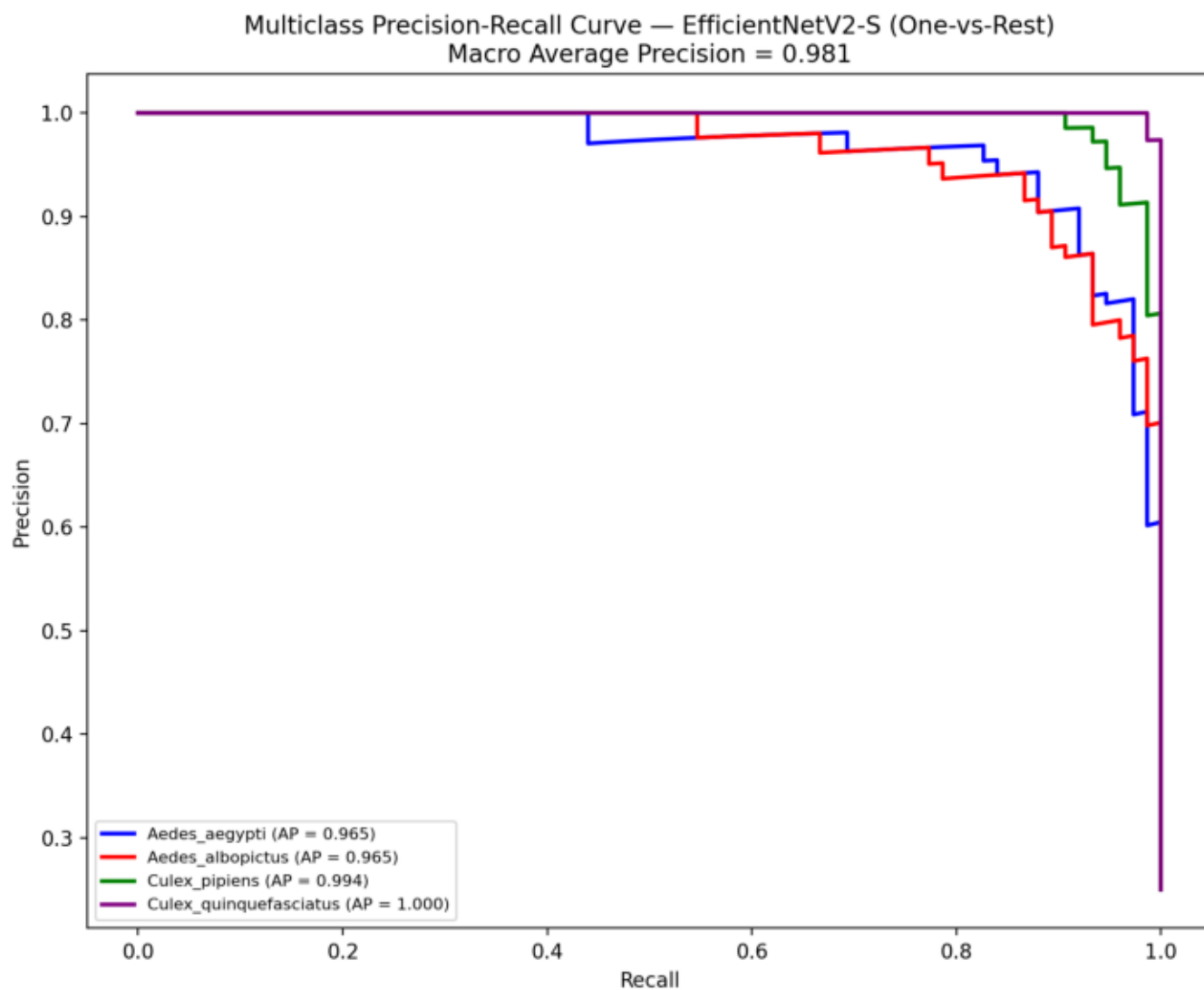
Confusion Matrix



Multiclass ROC Curve (One-vs-Rest)



Multiclass Precision-Recall Curve (One-vs-Rest)



Sample Predictions

True: *Culex quinquefasciatus*
Pred: *Culex quinquefasciatus*



True: *Aedes aegypti*
Pred: *Aedes aegypti*



True: *Aedes aegypti*
Pred: *Aedes aegypti*



True: *Culex pipiens*
Pred: *Culex pipiens*



True: *Aedes albopictus*
Pred: *Aedes albopictus*



True: *Culex quinquefasciatus*
Pred: *Culex quinquefasciatus*



True: *Aedes albopictus*
Pred: *Aedes albopictus*



True: *Culex quinquefasciatus*
Pred: *Culex quinquefasciatus*



True: *Aedes albopictus*
Pred: *Aedes albopictus*



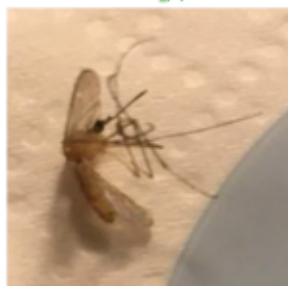
True: *Aedes aegypti*
Pred: *Aedes aegypti*



True: *Aedes aegypti*
Pred: *Aedes aegypti*



True: *Culex pipiens*
Pred: *Culex pipiens*



True: *Culex pipiens*
Pred: *Culex pipiens*



True: *Aedes albopictus*
Pred: *Aedes albopictus*



True: *Aedes albopictus*
Pred: *Aedes albopictus*



True: *Culex pipiens*
Pred: *Culex pipiens*



Final Results

Model Name: EfficientNetV2-S

Training Accuracy: 96.14%

Validation Accuracy: 92.00%

Testing Accuracy: 92.00%

Precision: 92.05%

Recall: 92.00%

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